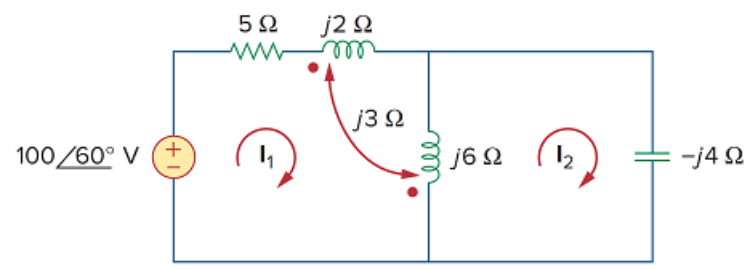


Problem

Determine the phasor currents I_1 and I_2 in the circuit of Fig. 13.13.

Figure 13.13



Step-by-step solution

Step 1 of 4

Refer to Figure 13.13 in the textbook.

Draw the circuit diagram with loop currents.

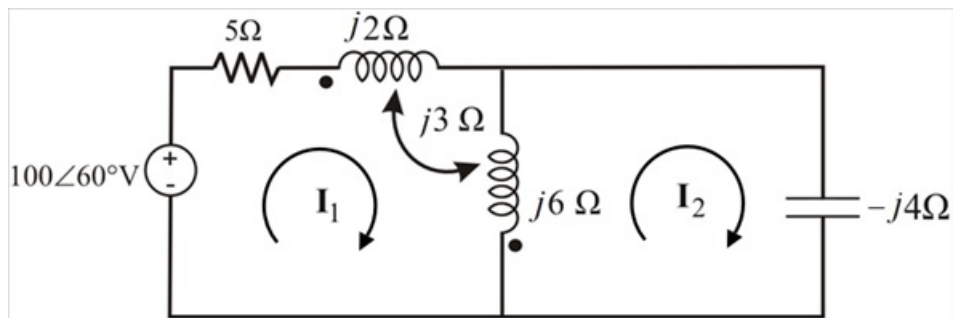


Figure 1

Step 2 of 4

Apply Kirchhoff's Voltage Law to loop 1.

$$-100\angle 60^\circ + 5I_1 + [j2I_1 - j3(I_1 - I_2)] + [j6(I_1 - I_2) - j3I_1] = 0$$

$$I_1(5 + j2 - j3 + j6 - j3) + I_2(j3 - j6) = 0$$

$$(5 + j2)I_1 - (j3)I_2 = 100\angle 60^\circ \quad \dots\dots (1)$$

Apply Kirchhoff's Voltage Law to loop 2.

$$-j4I_2 + [j6(I_2 - I_1) + j3I_1] = 0$$

$$(j3 - j6)I_1 + (j6 - j4)I_2 = 0$$

$$(j2)I_2 = (j3)I_1$$

$$I_2 = 1.5I_1$$

Step 3 of 4

Substitute $1.5\mathbf{I}_1$ for $\mathbf{I}_2$ in equation (1).

$$(5 + j2)\mathbf{I}_1 - (j3)(1.5\mathbf{I}_1) = 100\angle 60^\circ$$

$$(5 + j2)\mathbf{I}_1 - j4.5\mathbf{I}_1 = 100\angle 60^\circ$$

$$\begin{aligned}\mathbf{I}_1 &= \frac{100\angle 60^\circ}{5 - j2.5} \\ &= \frac{100\angle 60^\circ}{5.59\angle -26.565^\circ} \\ &= 17.889\angle 86.57^\circ \text{ A}\end{aligned}$$

Therefore, the value of the current, $\mathbf{I}_1$ is $\boxed{17.889\angle 86.57^\circ \text{ A}}$.

Step 4 of 4

Calculate the value of the current, $\mathbf{I}_2$.

$$\mathbf{I}_2 = 1.5\mathbf{I}_1$$

Substitute $17.889\angle 86.57^\circ \text{ A}$ for $\mathbf{I}_1$.

$$\begin{aligned}\mathbf{I}_2 &= 1.5(17.89\angle 86.57^\circ) \\ &= 26.83\angle 86.57^\circ \text{ A}\end{aligned}$$

Therefore, the value of the current, $\mathbf{I}_2$ is $\boxed{26.83\angle 86.57^\circ \text{ A}}$.